

GUJARAT TECHNOLOGICAL UNIVERSITY**BE - SEMESTER-V (NEW) EXAMINATION – WINTER 2021****Subject Code:3150912****Date:27/12/2021****Subject Name:Signals and Systems****Time:02:30 PM TO 05:00 PM****Total Marks: 70****Instructions:**

1. Attempt all questions.
2. Make suitable assumptions wherever necessary.
3. Figures to the right indicate full marks.
4. Simple and non-programmable scientific calculators are allowed.

		MARKS
Q.1	(a) Give expression of trigonometric Fourier Series. Define coefficients of DC, fundamental and harmonics.	03
	(b) Check whether the signal $x(t)=2\cos(100\pi t)+5\sin(50t)$ is periodic or not.	04
	(c) Obtain complex exponential Fourier series for $\sin \omega t$ and $\cos \omega t$. Assume ω to be constant.	07
Q.2	(a) Find even part of signal $x(n)=u(n)-u(n-4)$.	03
	(b) Define energy signal. Find the energy of the signal $x(t)=5e^{-4t}u(t)$	04
	(c) Check whether the system described by the equation $y(t)=10x(t)+5$ is linear, static, time invariant, causal and stable.	07
	OR	
	(c) Check whether the system described by the equation $y(n)=x(n)u(n)$ is linear, static, time invariant, causal and stable.	07
Q.3	(a) Give the convolution sum and integral formulas.	03
	(b) List and prove any two properties of convolution sum.	04
	(c) Convolve $x(t)=e^{-3t}[u(t)-u(t-2)]$ and $h(t)=e^{-t}u(t)$.	07
	OR	
Q.3	(a) State the linearity and time shifting property of Fourier transform.	03
	(b) Prove convolution property of Fourier transform.	04
	(c) Find inverse Fourier transform of	07
	$X(\omega) = \frac{2j\omega}{(2+j\omega)^2}$	
Q.4	(a) Define Z-transform. What is ROC?	03
	(b) Find the Z-transform of $4^n u(n)$.	04
	(c) Find inverse Z-transform of	07
	$X(z) = \frac{(z+4)}{(z^2-4z+3)}$	
	Assume right-sided sequence.	
	OR	
Q.4	(a) State any three properties of ROC of Z-transform.	03
	(b) State and prove the time shifting property of the Z-transform.	04
	(c) Solve the difference equation $y(n)-0.5y(n-1) = \delta(n)$ using Z-transform.	07
Q.5	(a) What is zero-order hold in sampling?	03

- (b) What are the effects of under sampling of a signal ? **04**
(c) Describe various types of sensors used for IoT applications. **07**

OR

- Q.5** (a) Explain the reconstruction of a signal from its samples. **03**
(b) Explain any one practical application of Signals and Systems theory. **04**
(c) Explain working of any system based on Arduino. **07**
